

# SIMATS ENGINEERING

Department of Computer Science and Engineering



## CSA15 Cloud Computing and Big Data Analytics CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense

Editable student template for application-based cloud virtualization and SDDC/cloud operations defense

### Assessment Metadata

|                 |                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------|
| Course Code     | CSA15                                                                                                                         |
| Course Title    | Cloud Computing and Big Data Analytics                                                                                        |
| Assessment      | CO2 AT3 - Capstone Cloud Infrastructure Case Analysis and Defense                                                             |
| Submission Type | Individual digital case analysis and infrastructure defense based on the same capstone title used in CO1, CO2 AT1 and CO2 AT2 |
| Faculty         | Dr C. Rajesh Babu                                                                                                             |
| Employee ID     | SSETSCS415                                                                                                                    |
| Submission Mode | Completed DOCX template and GitHub repository link through Google Classroom                                                   |

### How This Template Works

- Use the same capstone title used in CO1, CO2 AT1 and CO2 AT2.
- Do not recalculate all VM sizing or repeat all configuration screenshots. Summarize the final AT1/AT2 evidence and use it for design defense.
- Answer all five questions with application-oriented reasoning connected to the capstone product. • Include sustainable cloud operation decisions wherever relevant: right-sized resources, log retention, storage lifecycle, scheduled shutdown, API-call reduction and VM/container/serverless comparison.
- Paste the final topology diagram in the given response area. The sample diagram is only a structure; redraw it for the student's own product.
- Upload this completed document and supporting evidence to GitHub, then submit the repository link through Google Classroom.

#### Assessment Focus

AT1 answered what VM is required. AT2 showed configuration or simulation evidence. AT3 asks whether the student can defend the complete infrastructure design under application, security, operations and failure conditions.

### Student and Capstone Details

|                 |                                                                            |
|-----------------|----------------------------------------------------------------------------|
| Student Name    | Karanam Reddy Vishnu Chaithanya                                            |
| Register Number | 192372057                                                                  |
| Class / Slot    | Slot B                                                                     |
| Capstone Title  | SentinelX: Cloud Threat Intelligence Using NLP and Serverless Risk Scoring |

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|---------------------------|---------------------------------------------------------------------------------|
| CO1 Evidence Link         | [Insert CO1 concept map / presentation link from Google Classroom]              |
| CO2 AT1 VM Recommendation | vCPU: 4 RAM: 8 GB Storage: 50 GB VM Class: General Purpose (Burstable/Standard) |
| CO2 AT2 Evidence Link     | [Insert CO2 AT2 evidence link from Google Classroom]                            |
| GitHub Repository Link    | [Insert GitHub repository link for SentinelX]                                   |
| Submission Date           | [Insert submission date]                                                        |

## Evidence Carry-Forward Summary

|                                |                                                                                                                                                                     |                                                                                                                              |
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|                                |                                                                                                                                                                     |                                                                                                                              |
| CO1 concept map / presentation | CO1 concept map presented SentinelX as a cloud threat intelligence pipeline: ingest raw threat feeds, parse them with NLP, and score risk for analysts.             | Confirms the same problem statement (threat-intel triage) and analyst/admin roles carried into the AT3 topology and defense. |
| CO2 AT1 sizing workbook        | AT1 sizing workbook recommended a 4 vCPU / 8 GB RAM / 50 GB SSD general-purpose VM class as the baseline compute reference for the NLP/scoring workload.            | Baseline sizing is reused in Q1 and Q5 to justify right-sizing and cost/utilization decisions for the scoring engine.        |
| VM calibration sheet           | Calibration sheet benchmarked NLP parsing throughput and concurrent risk-scoring requests against the recommended VM class.                                         | Evidence that the chosen configuration can sustain expected ingest-and-score latency during a feed spike.                    |
| CO2 AT2 practical evidence     | AT2 practical evidence included cloud project configuration, serverless function deployment logs and dashboard hosting screenshots.                                 | Used in Q2 to justify template, snapshot and rollback decisions without repeating screenshots.                               |
| Capstone architecture decision | Serverless-first architecture: React dashboard calling an API Gateway/Cloud Functions layer that fronts the NLP engine, risk-scoring engine, database and alerting. | Forms the backbone of the Q3 topology diagram and Q4 deployment-alternative comparison.                                      |

## Q1: Virtualization Value and Capstone Cloud Deployment Summary

Explain your capstone product, expected workload, cloud need and deployment context. Analyze how virtualization improves cost efficiency, resource utilization, workload isolation, infrastructure flexibility and cloud deployment readiness for your product.

|                            |                                                                                                                                                                                                             |
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|                            |                                                                                                                                                                                                             |
| Capstone product and users | SentinelX is a cloud threat-intelligence platform that ingests threat feeds (CVE/IOC/OSINT text), uses NLP to extract indicators, and computes a serverless risk score for each event. Users are two roles: |

|                                   |                                                                                                                                                                                                                                                                                      |
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|                                   | security analysts who review scored alerts on a dashboard, and admins/SOC leads who monitor system health and configure feeds.                                                                                                                                                       |
| Expected workload                 | Bursty, event-driven workload: threat feeds arrive on a schedule or in irregular bursts (new CVE disclosures, breaking OSINT reports), each triggering an NLP parse and a risk-scoring computation; between feed updates the system is largely idle.                                 |
| Cloud need                        | Needs elastic compute that can absorb sudden parsing/scoring bursts without a permanently reserved server, reliable managed storage for raw threat text and structured risk records, and low latency alerting when a high-risk score is produced.                                    |
| Virtualization value              | Virtualization/serverless abstraction (Cloud Functions on managed, virtualized compute) removes the need to provision or patch a dedicated NLP/scoring server, letting the team focus on detection logic while the platform virtualizes compute per invocation.                      |
| Cost and utilization benefit      | Because threat-feed bursts are infrequent relative to idle time, serverless compute bills only for actual NLP/scoring invocations, avoiding a continuously running VM sized for worst-case ingest volume — directly improving utilization and cost efficiency.                       |
| Isolation and flexibility benefit | Each service — NLP parsing, risk scoring, authentication, storage, notification — runs in an isolated virtualized execution environment, so a spike or bug in the NLP engine cannot crash authentication or the analyst dashboard; new detection modules can be added independently. |

Case Analysis Response: SentinelX's value depends on infrastructure that stays idle between threat-feed updates and then processes a burst of text-parsing and risk-scoring work within seconds of a new disclosure. A dedicated VM sized for peak feed volume would sit mostly idle, wasting compute budget. Virtualization solves this at two levels: the cloud provider's infrastructure runs the NLP and scoring functions on virtualized, multi-tenant compute so SentinelX never owns idle hardware, and the AT1 sizing exercise (4 vCPU / 8 GB RAM / 50 GB SSD, general-purpose class) gives a concrete right-sizing baseline that the serverless auto-scaler treats as a reference ceiling rather than a fixed allocation. This directly improves cost efficiency, since SentinelX is billed per invocation instead of per hour of a running VM. Resource utilization improves because compute is consumed only for real events — feed ingest, NLP entity extraction, risk-score computation, alert dispatch — instead of being reserved continuously. Workload isolation is achieved because authentication, database, storage, NLP parsing and risk scoring each run as separately virtualized functions, so a burst in feed ingestion does not degrade the analyst dashboard or admin console. Infrastructure flexibility comes from being able to scale or replace any one service — for example, swapping the NLP model — without re-architecting the rest. Together these properties make SentinelX cloud-deployment-ready: it can move from a single-developer prototype to a continuously monitoring SOC tool by scaling the same virtualized building blocks rather than re-platforming.

## Q2: Rapid Provisioning, Testing and Virtual Machine Lifecycle Case

Using your CO2 AT1 VM recommendation and CO2 AT2 practical evidence, explain how virtual machines, VM images, templates, snapshots, clones or containers support rapid provisioning, testing, deployment, rollback and maintenance for your capstone product.

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| VM image or base environment | AT1 base VM image spec (4 vCPU / 8 GB RAM, general-purpose Linux image) used for local NLP-library testing and dependency setup.              | The same base runtime configuration is replicated as the Cloud Functions Node.js/Python runtime container, ensuring dev/test/prod parity for the NLP pipeline. |
| Template or repeatable setup | AT2 practical evidence: project template with pre-configured API routes, database schema and Cloud Functions triggers for ingest and scoring. | New environments (staging vs production) are provisioned from this template in minutes, avoiding manual reconfiguration of ingest and scoring routes.          |
| Snapshot / rollback          | AT2 captured configuration snapshots (function versions, database schema) before each deployment.                                             | Function versioning and schema history let the team roll back a bad risk-scoring deployment within minutes if it mis-scores threats.                           |
| Clone / testing copy         | AT2 cloned the production project into a staging environment for risk scoring load tests using sample threat feeds.                           | Testing the risk-scoring logic on a clone avoids risking live analyst-facing data during experiments with new scoring weights.                                 |

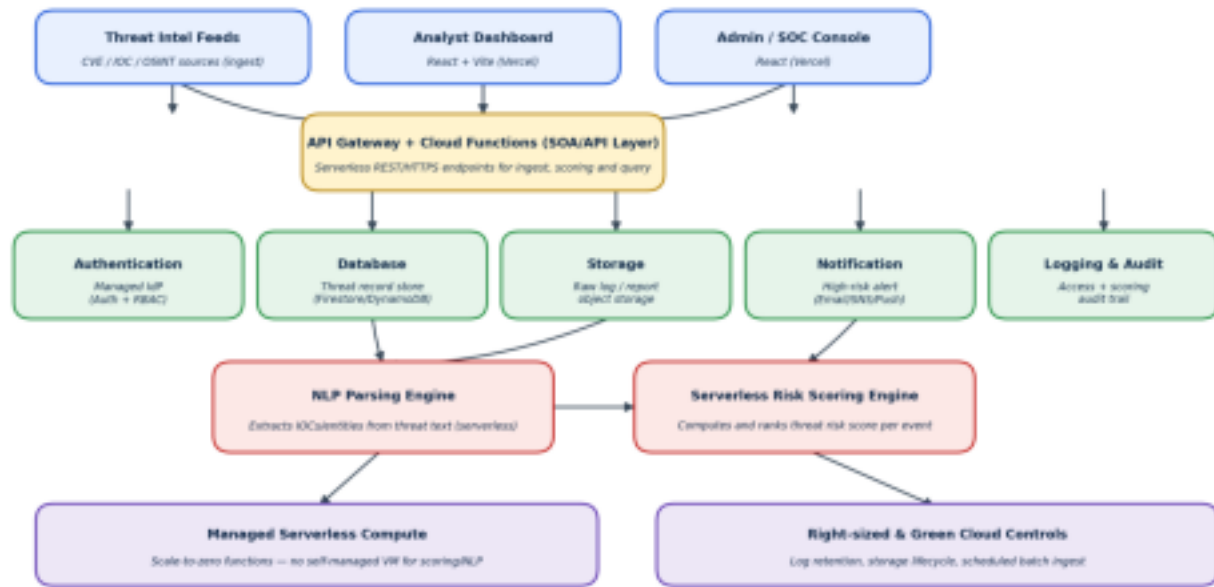
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| Container or VM decision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AT1 compared VM vs container vs serverless for the NLP/scoring backend and selected serverless (Cloud Functions) over a dedicated VM. | Confirmed in AT3: serverless functions match SentinelX's bursty, feed-driven workload far better than a continuously running VM. |
| Maintenance and update plan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | AT2 evidence of NLP library and dependency updates applied to the parsing and scoring functions.                                      | Scheduled monthly dependency and NLP model updates are applied and redeployed function-by-function without full-system downtime. |
| <p><b>Case Analysis Response: Lifecycle practices reduce both delay and risk in SentinelX because every stage — base image, template, snapshot, clone and container decision — is reusable and independently verifiable before it touches production. Using a fixed base runtime image for the NLP/scoring functions means a new contributor can reproduce the exact AT1/AT2 environment locally instead of debugging library-version drift, cutting setup delay from hours to minutes. The project template captures API routes, database schema and trigger bindings, so spinning up a new staging environment for a feature like a revised scoring model takes minutes rather than a manual reconfiguration that risks missing a route or permission. Snapshots of function versions and schema give an immediate rollback path: if a scoring deployment misclassifies threats, the previous version is restored without waiting for a fresh build, minimizing the window during which analysts see incorrect risk scores. Cloning production into a staging copy lets new NLP models or scoring weights be load-tested against realistic feed data without risking live analyst workflows. Finally, choosing serverless functions over a dedicated VM means routine maintenance — dependency bumps, NLP library patches — is applied per-function and rolled out gradually, rather than requiring a maintenance window that would take the whole scoring pipeline offline.</b></p> |                                                                                                                                       |                                                                                                                                  |

### Q3: Final Cloud Deployment Topology and SOA/API Service Integration Case

Draw and explain the final deployment topology for your capstone product. Show frontend, backend/API, database, storage, authentication, notification, analytics and any VM/container/managed cloud service components. Explain how SOA/API-based integration connects these modules and supports modular cloud service design.

## SentinelX — Cloud Deployment Topology

Cloud Threat Intelligence Using NLP and Serverless Risk Scoring



|                |                                                                                            | VM / Container / Managed Service /                  |                                                                                             |
|----------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------|
| Frontend       | React + Vite analyst dashboard and admin/SOC console                                       | Managed/serverless — static hosting + edge (Vercel) | HTTPS calls to API Gateway/Cloud Functions endpoints                                        |
| Backend/API    | API Gateway + Cloud Functions (Node.js/Python) for ingest, NLP and scoring endpoints       | Serverless (fully managed, no VM)                   | Exposes REST/HTTPS callable functions consumed by dashboard and console                     |
| Database       | Managed NoSQL/document store (Firestore/DynamoDB class) for threat records and risk scores | Managed/serverless NoSQL (SaaS)                     | Cloud Functions read/write via SDK; dashboard subscribes to live risk-score updates         |
| Storage        | Managed object storage for raw threat-feed text and generated reports                      | Managed object storage (SaaS)                       | Raw feed text and reports uploaded/retrieved via signed URLs referenced in database records |
| Authentication | Managed Authentication service with token-based sign-in                                    | Managed identity service (SaaS)                     | Clients authenticate via Auth SDK; Cloud Functions verify tokens on each call               |
| Notification   | Managed push/email                                                                         | Managed push/notification                           | Cloud Functions trigger a                                                                   |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                |                                          |                                                                                                        |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | notification service (FCM/SNS-class) for high-risk alerts                      | service (SaaS)                           | push/email alert when a new record crosses the high-risk threshold                                     |
| Analytics / AI module                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Serverless NLP parsing + risk scoring engine (custom logic on Cloud Functions) | Serverless (custom logic, scale to-zero) | Triggered by database onCreate events on new feed entries; calls storage then notification in sequence |
| Paste or draw the final capstone deployment topology diagram here.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                |                                          |                                                                                                        |
| <p>SOA/API Integration Explanation: SentinelIX's modules communicate exclusively through well-defined HTTPS/callable APIs and database event triggers rather than direct database access from the client, which is the core of its SOA-style design. When a new threat-feed entry arrives, an ingest Cloud Function validates and writes it to the database; a database onCreate trigger fires the NLP parsing function, which extracts entities/IOCs and hands them to the risk-scoring function, which computes a score and — if it crosses a threshold — invokes the notification service. Each of these — Auth, Database, NLP parsing, Risk scoring, Notification, Storage — is an independently deployable, independently scalable service exposed through its own API surface. This modular design means the NLP model or scoring algorithm can be swapped or retrained without touching the dashboard or authentication code, and a failure in one module (for example, notification rate-limiting) degrades only that capability instead of the whole platform. It also lets the analyst dashboard and admin console reuse the exact same backend API surface instead of duplicating business logic.</p> |                                                                                |                                          |                                                                                                        |

Q4: Design Alternatives, Identity, Access and Federated Cloud Privacy Case

Compare at least two deployment alternatives for your capstone product, such as single VM, multi-tier VM, containerized deployment, managed backend, federated cloud or hybrid cloud model. Justify the selected design by analyzing identity management, role-based access control, federated access, data sharing, privacy protection and security risks.

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| Alternative 1 | Simple to reason about; one deployment target; predictable networking. | Cannot scale NLP parsing and risk scoring independently; a crash or memory spike in the NLP engine can take down authentication and ingest together; | All roles typically share one access surface, making fine-grained RBAC harder to enforce and increasing the blast radius of a credential leak. |
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|                           |                                                                                                                                                         | requires manual OS patching and idle-hour cost.                                                                                                                                                          |                                                                                                                                                                                                                                             |
| Alternative 2             | Each module (auth, NLP, scoring, notification) runs in its own container; independent scaling and restart; portable across cloud providers.             | Higher operational complexity (orchestration, networking, secrets management) than a small student team can realistically maintain; still needs always-on cluster nodes, reducing the idle-cost benefit. | Supports per-service identity and network policies, but secrets/service account sprawl across containers raises federated-access risk if not centrally managed.                                                                             |
| Selected design           | Matches SentinelX's bursty, feed-driven workload; pay-per invocation cost; each function isolated and independently scalable; no OS/VM patching burden. | Vendor lock-in to the chosen cloud provider; cold-start latency on rarely-used functions; less low-level control over networking.                                                                        | Identity is centralized with per-user tokens verified on every function call, and database security rules enforce role based access (analyst vs admin) at the data layer — the smallest practical blast radius for a student-team capstone. |
|                           |                                                                                                                                                         |                                                                                                                                                                                                          |                                                                                                                                                                                                                                             |
| User roles                |                                                                                                                                                         | Analyst (reviews scored alerts) and Admin/SOC lead (configures feeds, monitors system) — enforced via a role field on the authenticated user's profile record.                                           |                                                                                                                                                                                                                                             |
| Authentication method     |                                                                                                                                                         | Managed identity provider with email/password or SSO-based sign in and verified tokens (no locally stored passwords).                                                                                    |                                                                                                                                                                                                                                             |
| Role-based access control |                                                                                                                                                         | Database security rules check the authenticated user's role claim before allowing reads/writes to feed, scoring or                                                                                       |                                                                                                                                                                                                                                             |



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|                                                                                                                                                                                                                                                                                                                                                                                  | admin-config collections.                                                                                                                                                                                   |
| Federated access / partner access                                                                                                                                                                                                                                                                                                                                                | No third-party/partner federation in the current build; future integration with external SOC/SIEM tools is scoped as a federated read-only API with its own service-account credentials, not shared logins. |
| Sensitive data handled                                                                                                                                                                                                                                                                                                                                                           | Raw threat-feed text (which may reference real organizations/IPs), extracted IOCs, computed risk scores and analyst review notes — classified as sensitive security data.                                   |
| Privacy protection decision                                                                                                                                                                                                                                                                                                                                                      | Feed source details and raw text are only exposed to authenticated analyst/admin roles, never publicly; database rules deny cross-role reads by default and access is logged for audit.                     |
| <b>Case Analysis Response: The serverless/managed-service design is defended over both alternatives primarily on identity and privacy grounds appropriate to a threat-intelligence product handling sensitive security data. A single monolithic VM would force all roles to authenticate against one broad surface, making it hard to guarantee that an analyst account can</b> |                                                                                                                                                                                                             |

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never read admin-only configuration or another team's private feed sources — a real risk for a security tool. The containerized alternative solves isolation but multiplies the number of service accounts, secrets and network policies a small student team must manage correctly, and a single misconfigured container-to-container link becomes a plausible leak path for threat data. The selected serverless design centralizes identity in a managed Auth service, enforces RBAC at the data layer through database security rules rather than trusting application code, and keeps the attack surface to well audited managed APIs instead of self-managed networking. This trades some vendor flexibility for a materially smaller privacy and security blast radius, which is the right trade-off for a system whose core purpose is protecting sensitive threat-intelligence data.

## Q5: Software-Defined Datacenter and Cloud Operations Defense Case

Analyze how your infrastructure will respond to operational and failure scenarios such as VM failure, database growth, traffic spike, credential leakage, service outage, resource contention or cost increase. Explain how SDDC concepts and cloud operations practices such as automated provisioning, software-defined networking, software-defined storage, monitoring, logging, backup, rollback, scaling, resource control and maintenance improve the final infrastructure design. Include Green Computing decisions such as resource right-sizing, storage lifecycle control, log retention, API-call reduction, scheduled shutdown, VM/container/serverless comparison and carbon/energy proxy indicators where relevant.

|            |                                                                                                                                                         |                                                                                                                                                                    |                                                                                                                  |
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|            |                                                                                                                                                         |                                                                                                                                                                    |                                                                                                                  |
| VM failure | Since the backend runs as serverless functions rather than a single VM, a 'VM failure' equivalent is a Cloud Functions region outage — feed ingest, NLP | The managed cloud provider automatically reschedules function instances across its infrastructure; multi-zone database replication keeps threat records available. | Cloud provider status dashboard and Cloud Functions retry/error logs monitored for repeated invocation failures. |

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|                    | parsing and scoring pause.                                                                                                         |                                                                                                                                                                     |                                                                                                       |
| Database growth    | Database record count grows as ingested feed entries, extracted IOCs and risk scores accumulate, raising cost and query latency.   | Composite indexes on frequent queries (recent high-risk items, feed source) and scheduled archival of resolved/old feed entries to cold storage.                    | Database usage dashboard; scheduled function archives entries older than a defined retention window.  |
| Traffic spike      | A major CVE disclosure or breaking OSINT event can trigger a burst of simultaneous feed entries requiring NLP parsing and scoring. | Cloud Functions auto-scale horizontally per invocation; the managed database's distributed architecture absorbs concurrent writes without manual capacity planning. | AT1 calibration sheet load numbers used as the baseline concurrency the system is expected to absorb. |
| Credential leakage | A leaked API key or service-account key could allow unauthorized reads of raw threat data or risk scoring results.                 | Database security rules (not client-side checks) gate all data access; service-account keys are never bundled in client code and are rotated on suspicion of leak.  | Auth sign-in logs and database rule-denial logs reviewed for anomalous access patterns.               |
| Service outage     | Loss of the notification path would mean                                                                                           | Notification delivery failures fall back to in-app polling of the                                                                                                   | Cloud Function logs on notification send failures; alerting                                           |

|                     |                                                                                                                      |                                                                                                                |                                                                                     |
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|                     | analysts stop receiving high-risk alerts even though scores are still being computed.                                | database so the dashboard still surfaces new high-risk items.                                                  | on send-error rate.                                                                 |
| Resource contention | Multiple functions writing to the same 'active feed' record concurrently could cause contention/hot document errors. | Restructure hot records into sharded sub-collections and use database transactions for the risk-scoring write. | Database contention error logs monitored; AT2 evidence of transaction-based writes. |

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| Cost increase                              | Unbounded function invocations or unarchived raw-feed storage could drive unexpected billing growth. | Right-sized function memory/timeout settings from AT1, storage lifecycle rules to auto-archive old raw-feed text, and billing alerts.                                          | Cloud budget alert configured at a defined monthly threshold; storage lifecycle policy evidence. |
|                                            |                                                                                                      |                                                                                                                                                                                |                                                                                                  |
| Automated provisioning                     |                                                                                                      | Project configuration (database rules, indexes, Cloud Functions) is version-controlled and deployed via CLI, so a new environment is provisioned from code, not manual clicks. |                                                                                                  |
| Software-defined networking                |                                                                                                      | Provider-managed HTTPS endpoints and edge network route traffic; no manually managed VLANs/subnets are required for this managed service architecture.                         |                                                                                                  |
| Software-defined storage                   |                                                                                                      | The document database and object storage are software-defined managed storage tiers, with lifecycle rules controlling retention programmatically.                              |                                                                                                  |
| Monitoring and logging                     |                                                                                                      | Cloud Functions logs and a performance-monitoring dashboard track invocation errors, latency and NLP/scoring failures.                                                         |                                                                                                  |
| Backup and rollback                        |                                                                                                      | Scheduled database exports for backup; Cloud Functions version history enables one-command rollback of a bad deployment.                                                       |                                                                                                  |
| Scaling and elasticity                     |                                                                                                      | Serverless auto-scaling handles feed-volume spikes without manual intervention; no pre-provisioned server capacity to size for peak.                                           |                                                                                                  |
| Resource control and maintenance           |                                                                                                      | Function memory/timeout limits set per AT1 sizing; dependency and NLP-model updates applied and redeployed function-by-function on a monthly cadence.                          |                                                                                                  |
| Green Computing and sustainability control |                                                                                                      | Scale-to-zero serverless compute means no energy is consumed during SentinelX's idle periods between feed updates.                                                             |                                                                                                  |
|                                            |                                                                                                      |                                                                                                                                                                                |                                                                                                  |
| VM right-sizing from CO2 AT1               |                                                                                                      | The 4 vCPU / 8 GB RAM baseline from AT1 is used as the function memory/CPU ceiling, avoiding over-provisioned compute that would                                               |                                                                                                  |

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|  | waste energy during idle periods between feed bursts. |
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| VM vs container vs serverless choice                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Serverless was chosen specifically because SentinelX's event-driven, low-duty-cycle ingest workload wastes the least energy when compute scales to zero between threat-feed updates, versus a VM or container that stays warm. |
| Scheduled shutdown / idle-resource control                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | No dedicated VM to shut down — functions naturally suspend after idle timeout; only the staging/testing environment is manually torn down after each testing session.                                                          |
| Log and file retention policy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Function logs retained 30 days; raw threat-feed text moved to cold storage after 90 days and deleted after the retention period required for audit purposes.                                                                   |
| Storage lifecycle and backup cleanup                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Storage lifecycle rules auto-transition or delete old raw-feed files; database backups older than the retention window are pruned.                                                                                             |
| API-call or polling reduction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Event-driven database triggers (push-based) are used instead of client polling for new feed entries, cutting redundant API calls from the dashboard.                                                                           |
| Carbon/energy proxy indicator to track                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Total Cloud Function invocation-seconds per month is tracked as a proxy for compute-energy use, alongside the cloud provider's own regional data-center efficiency reporting.                                                  |
| <b>Final Infrastructure Defense Statement: The SentinelX infrastructure is ready for pilot deployment, not yet full production. Reliability is defended by the managed provider's multi-zone infrastructure, function-level retry logic and database transactional writes for the contention-prone scoring path. Security is defended by centralizing identity in a managed Auth service and enforcing RBAC through database security rules rather than client-side checks, with no long lived secrets in client code. SDDC/cloud-operations maturity is shown through version-controlled, CLI-deployed infrastructure, function-level logging/monitoring and documented rollback via Cloud Functions versioning. Green Computing is addressed through scale-to-zero serverless compute, storage lifecycle rules and push- rather than poll based data sync. Before moving to production, the design still needs: load-testing at true multi-feed-source volumes, a documented incident-response runbook for credential leakage, and a formal SLA/backup-restore drill for the database — all flagged as next steps for the capstone's production readiness review.</b> |                                                                                                                                                                                                                                |

## CO2 Reflection and Individual Defense

This section must be individually written. It should show what the student understands from the complete CO2 chain: infrastructure sizing, practical configuration evidence and final operations defense.

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| What is the strongest infrastructure decision in your capstone design? | Choosing serverless Cloud Functions over a dedicated VM is the strongest decision — it matches SentinelX's bursty, feed-driven workload without paying for idle capacity, and each module (NLP, scoring, notification) scales and fails independently.                |
| Which CO2 concept became clearer after AT1, AT2 and AT3?               | Virtualization's role in workload isolation became far clearer after AT3 — AT1/AT2 focused on sizing a single VM, but defending the full topology showed why splitting auth, NLP parsing and scoring into isolated virtualized functions actually protects the system |

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|                                                                                                  | during failure.                                                                                                                                                                                           |
| Which risk is most serious for your capstone: failure, cost, privacy, scaling or sustainability? | Credential/data leakage is the most serious risk for SentinelX, because raw threat-feed data and risk scores are security-sensitive, and a leak could expose an organization's own vulnerability posture. |
| What evidence would you show during viva to defend your infrastructure design?                   | AT1 sizing workbook and calibration sheet, AT2 deployment screenshots, the AT3 topology diagram, and the database security rules file showing RBAC enforcement.                                           |
| What will you improve before final capstone deployment?                                          | Add automated load testing for the NLP/risk-scoring pipeline under simulated high-volume feed bursts, and formalize a credential rotation and incident-response runbook before any production pilot.      |

## CSA15 | Cloud Computing and Big Data Analytics CO2 AT3 | Student Template

## Student-Facing Assessment Criteria

|                                        |                                                                                                             |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------|
|                                        |                                                                                                             |
| Capstone continuity                    | The answer uses the same capstone title and connects CO1, CO2 AT1 and CO2 AT2 evidence.                     |
| Virtualization and lifecycle reasoning | VM, image, snapshot, clone, container or managed-service decisions are technically justified.               |
| Topology and integration clarity       | Architecture diagram and SOA/API explanation show clear module interaction.                                 |
| Identity, access and privacy           | Roles, authentication, RBAC, federated access, sensitive data and privacy decisions are defended.           |
| SDDC/cloud operations                  | Failure response, monitoring, logging, scaling, backup, rollback and resource control are addressed.        |
| Green Computing defense                | Right-sizing, retention, API efficiency, idle-resource control and sustainable cloud choices are explained. |
| Submission quality                     | GitHub, template completion, evidence references and individual reflection are complete.                    |

## GitHub and Google Classroom Submission

|                        |                                                                                                               |
|------------------------|---------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                               |
| README.md              | Capstone title, student details, final VM recommendation, deployment topology summary and AT3 evidence index. |
| Completed AT3 document | This completed editable DOCX or exported PDF if instructed.                                                   |
| Diagram image          | Final topology diagram if created separately.                                                                 |

|                             |                                                                                           |
|-----------------------------|-------------------------------------------------------------------------------------------|
| <b>Evidence references</b>  | Links or screenshots from CO1, CO2 AT1 and CO2 AT2 as needed.                             |
| <b>Green Computing note</b> | Short README section explaining sustainable cloud choices and resource-control decisions. |
| <b>Google Classroom</b>     | Submit the GitHub repository link through GCR.                                            |

## Student Assurance

I confirm that this case analysis, design reasoning, topology diagram and final defense statement were prepared individually by me for the stated capstone title. I have not uploaded passwords, tokens, private keys or sensitive account information.

|                        |                                 |
|------------------------|---------------------------------|
| <b>Student Name</b>    | Karanam Reddy Vishnu Chaithanya |
| <b>Register Number</b> | 192372057                       |
| <b>Signature</b>       |                                 |
| <b>Date</b>            |                                 |

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## Faculty Verification

|                                        |                   |
|----------------------------------------|-------------------|
| <b>Faculty Name</b>                    | Dr C. Rajesh Babu |
| <b>Employee ID</b>                     | SSETSCS415        |
| <b>Constructive Feedback / Remarks</b> |                   |
| <b>Strength Observed</b>               |                   |
| <b>Improvement Suggested</b>           |                   |
| <b>Signature / Date</b>                |                   |

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